

Priya Abiram: Fluids Blade Design Project Overview

Your page must include:

- **Project overview:** What you were asked to design and why
- **Design process:** Key decisions, iterations, or modeling steps your team performed
- **Testing summary:** How the blades were tested and what data you analyzed
- **Your contribution:** Briefly describe what *you* worked on
- **Figures:** Plots, CAD image, and photos

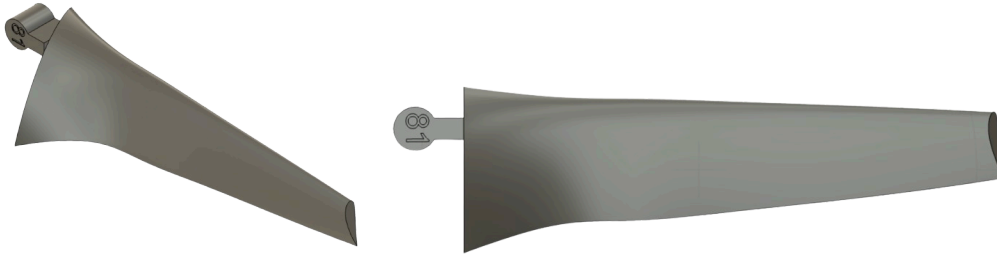
This is not a full technical report. Aim for 2–4 concise paragraphs plus 1–3 informative graphics.

This project aimed to design, fabricate, and test a 6-inch wind turbine blade that maximizes aerodynamic efficiency while satisfying geometric, structural, and manufacturability constraints. Using Blade Element Momentum Theory (BEMT), experimentally validated airfoil data, and an iterative design process, we developed a NACA 4412-based blade with smoothed chord and twist profiles, constrained by a 2-inch axial projection limit, 44 MPa flexural-stress limit, and 3D-printing requirements.

The design process included airfoil characterization at three Reynolds numbers, selection of an energy-weighted design wind speed, initialization of geometry, full-span BEMT convergence, tip and root smoothing, and structural verification. Predicted performance at 812 RPM suggested a power output of ~1 W. Experimental testing demonstrated peak power of 1.29 W near 2900 RPM, with the blade maintaining structural integrity and achieving $TSR \approx 6$: double the TSR it was originally designed for.

While all structural and geometric constraints were satisfied, several unexpected trends emerged, including the absence of a clear performance peak, resonance phenomena near 2100–2500 RPM, and a higher-than-anticipated operational TSR. Future iterations will increase design TSR, refine chord lengths near the outer span, and redesign geometry to avoid resonance-sensitive regions.

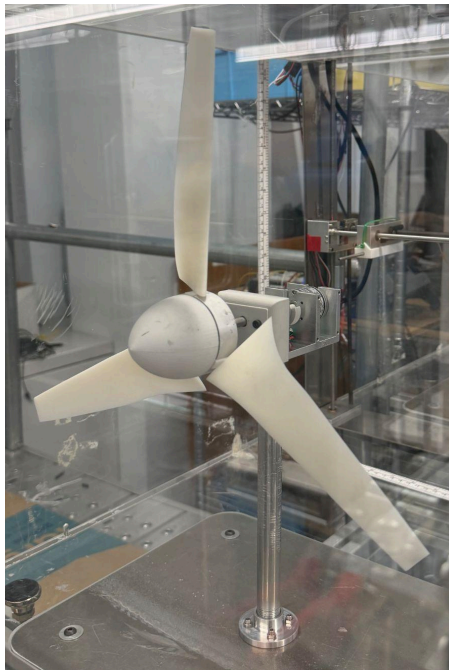
Blade final CAD design out of hub:



Blades in Hub:



Blade in wind tunnel:



This project required the design of a small horizontal-axis wind turbine blade for laboratory testing. The blade needed to operate efficiently within the wind-speed ranges created by the course's fan system and remain structurally safe under torque-brake loading. The

deliverables included a manufacturable 3D-printed blade and a performance evaluation based on experimental data.

These are the main aspects we wanted to optimize:

1. Maximize aerodynamic efficiency and total power extraction.
2. Ensure manufacturability using a 3D printer with 0.1 mm resolution.
3. Maintain smooth and continuous chord and twist geometry along the blade.
4. Ensure that the blade meets all geometric and structural constraints.

Here are the base requirements given to us:

1. Geometric: radius less than or equal to 6 inches, axial projection less than or equal to 2 inches, chord between 6 mm and 60 mm, use of a NACA 4412 airfoil.
2. Structural: flexural stress must remain below 44 MPa.
3. Aerodynamic: must perform well under wind speeds characterized by a Weibull distribution with given parameters. Used energy weighted velocity: 5.04 m/s
4. Manufacturing: blade must be printable without overhang failures and must retain continuous surface geometry.

Because wind power scales with U^3 , an energy-weighted average wind speed was computed using a Weibull distribution to define a representative design condition for blade optimization.

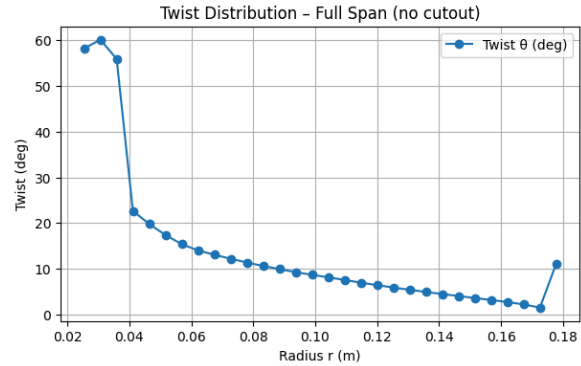
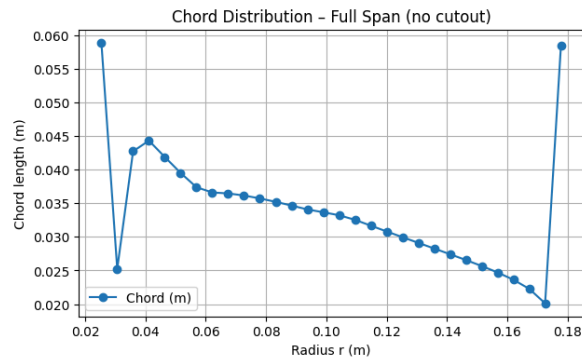
$U_{\text{energy}} = 5.04 \text{ m/s}$

We hypothesized that maintaining a relatively larger chord farther out along the blade would increase torque production due to higher local velocities. Pitch was selected near the optimal lift-generating angle without risking stall.

NACA 4412 lift and drag data were imported at Reynolds numbers of 50,000, 100,000, and 200,000. Interpolation allowed us to match airfoil characteristics to local flow conditions along the blade.

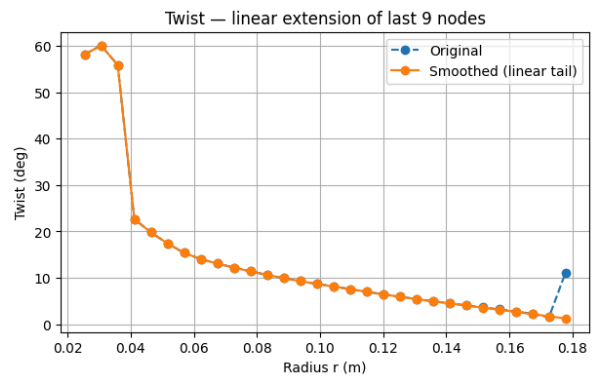
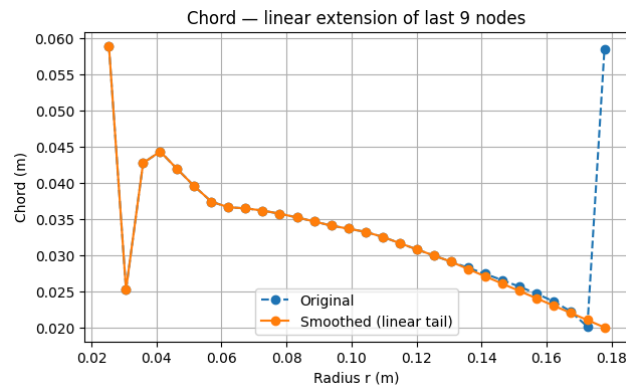
Blade Element Momentum Theory iterates for values of axial and tangential induction factors that cause our two predicted thrust and torque to converge. These induction factors are used for calculating the pitch and chord of each of our stations. A diagnostic BEMT run identified unstable behavior near the root and tip, especially within the inner 25 percent of the span where TSR becomes small. This guided later smoothing steps.

Thirty stations along the blade were solved iteratively to convergence for induction factors. The resulting chord and twist profiles contained numerical outliers at both ends of the blade.

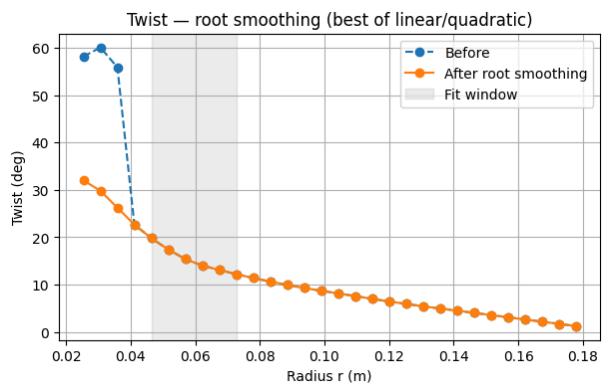
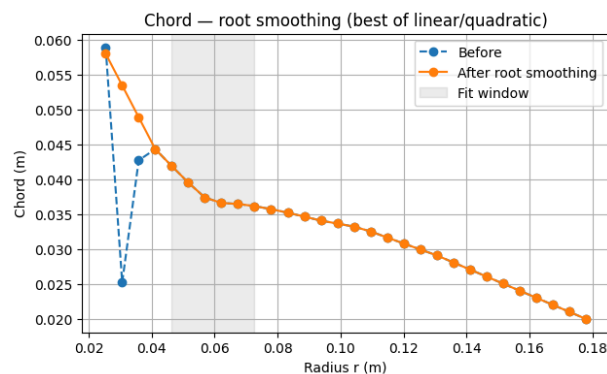


Tip smoothing was performed using a linear extrapolation of the final nine stable points. Root smoothing used a quadratic extrapolation of the closest stable upstream data. Both ensured smooth and manufacturable geometry while preserving aerodynamic fidelity.

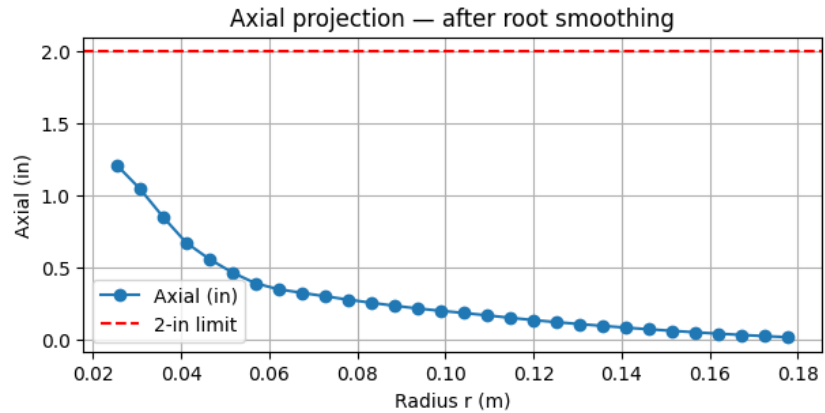
Tip Smoothing:



Root Smoothing:



We calculated axial projection at each station and confirmed it remained below the 2-inch limit after smoothing.



Performance Prediction: At our chosen Tip Speed Ratio and RPM (~812 RPM):
 Our predicted torque is 0.0117 Nm
 Our predicted power is 0.998 W

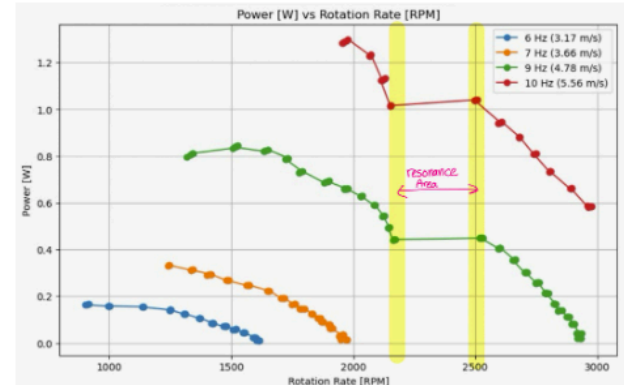
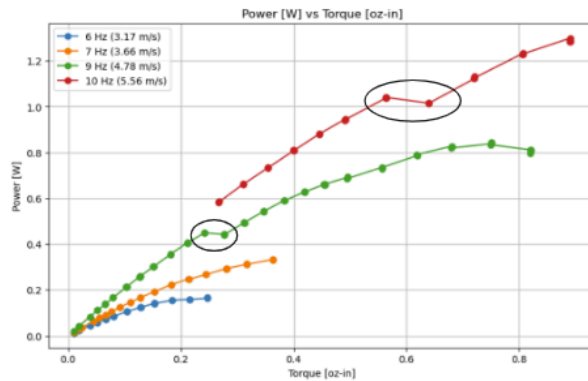
Using flapwise bending moment distribution and a rectangular-section stress approximation, we verified stresses remained well below the 44 MPa allowable limit.

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===== DESIGN-POINT TORQUE SUMMARY =====
Tip-Speed Ratio ( $\lambda_{\text{design}}$ ): 3.00
Angular speed  $\Omega_{\text{design}}$  : 85.04 rad/s (812 RPM)
Total torque  $Q_{\text{design}}$  : 0.0117 N·m
Power at design speed : 0.998 W

Torque rises with radius because outer sections:
• experience higher relative velocity → greater dynamic pressure, and
• act through a longer lever arm → larger moment contribution.
The final drop near the tip is geometric: as the Prandtl tip-loss
factor → 0, local loading vanishes even though radius is largest.
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Experimental Power Output

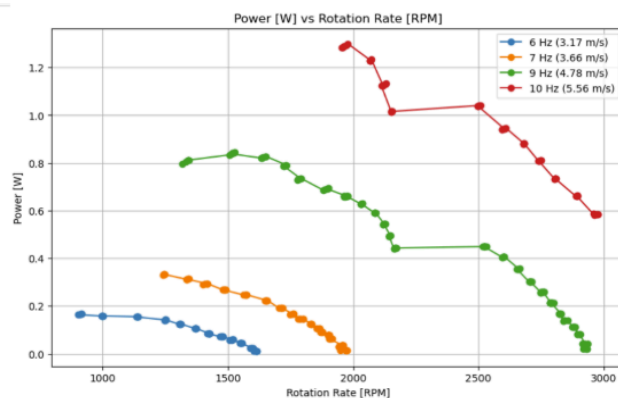
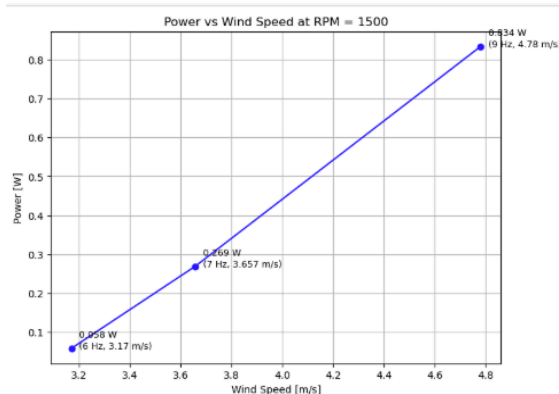
- Maximum measured power: 1.29 watts at 5.56 m/s
- Power increased consistently with wind speed, without showing a clear peak within the tested range
- A resonance region occurred between about 2100 and 2500 RPM, temporarily lowering measured power



Max Power = 1.29W at 5.56m/s

Although the blade was designed for $TSR = 3$, the experimental turbine consistently operated near $TSR = 6$. Despite this mismatch, the performance remained strong. This suggests the blade could be redesigned for a higher design TSR in future iterations. Predicted torque was $0.0117 \text{ N}\cdot\text{m}$, with a predicted power of approximately 1 watt at 812 RPM. The experimental blade exceeded this prediction because it operated at significantly higher RPM and TSR .

The blade exhibited no structural failure or undesirable deformation during testing. Flexural stresses remained well within the 44 MPa limit.



We successfully designed, manufactured, and tested a wind turbine blade that met all structural and geometric constraints and delivered higher-than-expected aerodynamic performance. The blade exceeded 1 watt of mechanical power output and operated stably across the wind-speed range. Although the turbine operated at a significantly higher TSR than originally designed, the overall experiment validated the robustness of the BEMT-based design process.